

2

Probability

People commonly talk about **chance** using everyday language. For example:

"St Mirren are likely to win this weekend."



"It is probably not going to rain today."



"The train might arrive at Glenfinnan Station on time."



"I'm pretty sure my cat is going to break the glass."



However, none of these statements seek to assign a **numerical** value to *quantify* those chances. In this chapter, probability as a *numerical measure* will be introduced, as well as related ways of presenting information.

2.1 Probability and Contingency Tables

In probability theory, an **event** is something which will either *happen or not* subject to *random chance*, such as:

Rolling a die and getting a five.

{ 1 2 3 4 5 6 }
 X X X X ✓ X

Tossing a coin and getting tails.

{ heads tails }
 X ✓

For any single *trial*, an event will either occur or not occur. The **probability** of an event is a numerical measure of how likely the event is to occur. Probability can be expressed as a fraction, a decimal or a percentage. Over a large number of repeated trials, the relative frequency of an event will often approach its probability. The probability of an event is written using P followed by a simple, understandable description of the event in brackets.

$$P(\text{five}) = \frac{1}{6}$$

$$P(\text{tails}) = \frac{1}{2}$$

In other words, if we roll a die a large number of times, we might expect to get a five for *approximately* a sixth of the rolls. However, due to random chance it may not happen *exactly* a sixth of the time. To obtain each of the probabilities above, the following formula can be used:

Probability of an Event:

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

For this formula to apply, the *outcomes* must be **equally likely**. This is the case, for example, when tossing a fair coin or rolling a fair die.

Example 1

A bag contains 7 red counters, 3 blue counters and 1 green counter. One counter is chosen at random, with each counter equally likely to be selected. State the probability that the counter chosen is:

- (a) Red. (b) Blue. (c) Green. (d) Yellow.

Given the probability of *success* for an event, $P(\text{event})$, the probability of *failure* (its **complement**) can be calculated:

$$P(\text{not event}) = 1 - P(\text{event})$$

Example 2

The UK has 76 cities. The table shows the number in each of England, Scotland, Wales and Northern Ireland.

Constituent country	England	Scotland	Wales	Northern Ireland
Number of cities	55	8	7	6

A city from the table is picked at random. State the probability that the city chosen is in:

- (a) Wales. (b) Not Wales. (c) France. (d) The UK.

A common table used to record frequencies relating to *two* variables is a **contingency table**.

When considering two events in probability, “*and*” means that *both* events happen, while “*or*” means that *at least one* of the events happens (including the possibility that they both happen, where applicable).

Example 3

The UK has 76 cities, most of which have a cathedral. The table shows the number of cities in each of England, Scotland, Wales and Northern Ireland, broken down by whether or not those cities have a cathedral.

	England	Scotland	Wales	Northern Ireland
Cathedral	43	6	7	5
No cathedral	12	2	0	1

A city from the table is picked at random. State:

- (a) $P(\text{cathedral})$ (b) $P(\text{Scotland})$
- (c) $P(\text{Scotland and cathedral})$ (d) $P(\text{Scotland or cathedral})$

Example 4

The table below shows the pupil numbers within one area of a school during a particular period, broken down by subject and course level.

	National 5	Higher
English	21	15
History	17	11

A pupil is selected at random to run an errand. State:

- (a) $P(\text{English})$ (b) $P(\text{Higher})$
- (c) $P(\text{English and Higher})$ (d) $P(\text{English or Higher})$

If an event has a probability of p , then we should *expect* it to happen approximately $n \times p$ times across n trials. However, since random chance is involved, we cannot *guarantee* how many times it will happen.

Example 5

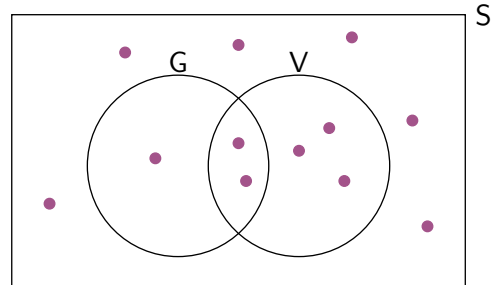
An insurance company knows that the probability one of the motorists it insures makes a claim during any given year is 15%. Find how many claims the company might expect to receive if it insures 27 000 motorists.

2.2 Venn Diagrams

Venn diagrams offer an alternative way of representing information to a contingency table. For instance, the contingency table below shows the twelve dishes from a restaurant's main course menu, broken down by whether or not each is *gluten-free* (G) and whether or not each is *vegan* (V). Beside it is a Venn diagram representing the same information. The letter S is sometimes used to represent *all possible observations*.

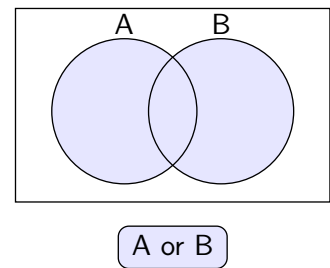
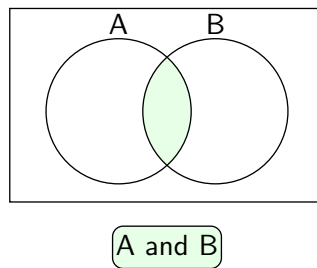
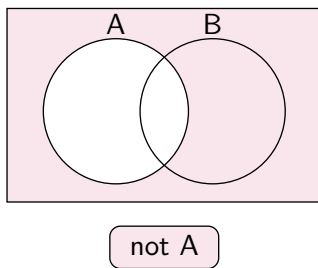
	vegan	not vegan
gluten-free	2	1
not gluten-free	3	6

Contingency table



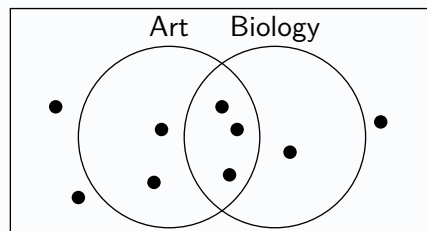
Venn diagram

Given two sets, such as A and B, it is important to be able to identify the following regions of a Venn diagram:



Example 1

A form tutor asks the nine pupils in their tutor group whether or not they are taking Art this year, and whether or not they are taking Biology this year. The results are shown in the Venn diagram below.

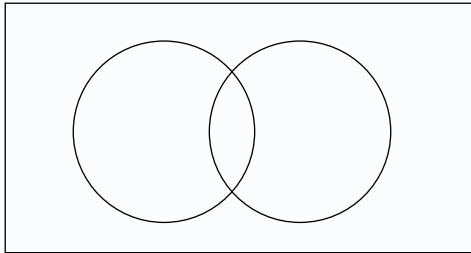


A pupil is selected at random from the tutor group. Find:

- | | | |
|-------------------------------------|-------------------------------------|---|
| (a) $P(\text{Art})$ | (b) $P(\text{Biology})$ | (c) $P(\text{Art and Biology})$ |
| (d) $P(\text{not Art})$ | (e) $P(\text{not Biology})$ | (f) $P(\text{Art or Biology})$ |
| (g) $P(\text{Art but not Biology})$ | (h) $P(\text{Biology but not Art})$ | (i) $P(\text{neither Art nor Biology})$ |

Example 2

100 pupils are asked whether or not they like cats, and whether or not they like dogs, with the results shown in the table below. Represent this information on the Venn diagram below.



	like cats	do not like cats
like dogs	34	12
do not like dogs	45	9

A pupil is selected at random. Find:

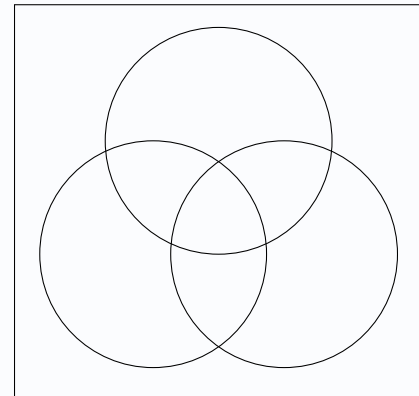
- (a) $P(\text{likes cats})$ (b) $P(\text{likes dogs but not cats})$ (c) $P(\text{likes cats but not dogs})$

Constructing Venn diagrams and using them to calculate probabilities can be extended to problems with three groups.

Example 3

A group of 160 college students were asked which study facilities they use from a choice of the library, the study café and the computer suite. The results were as follows:

- 4 students use all three facilities.
- 38 students use both the library and the study café.
- 32 students use both the library and the computer suite.
- 15 students use both the study café and the computer suite.
- 90 students use the library.
- 85 students use the study café.
- 53 students use the computer suite.
- 13 students use none of these facilities.



The numbers of students who are listed as using two facilities include those who use all three facilities.

- (a) Complete the Venn diagram to show this information.
- (b) One of the college students is selected at random. Determine the probability that the selected student uses the study café and the computer suite, but not the library.

2.3 Tree Diagrams

If knowing that one event has happened does not change the probability that another event will happen, then those two events are said to be **independent**. If two events A and B are independent then the probability of *both* events happening can be calculated as:

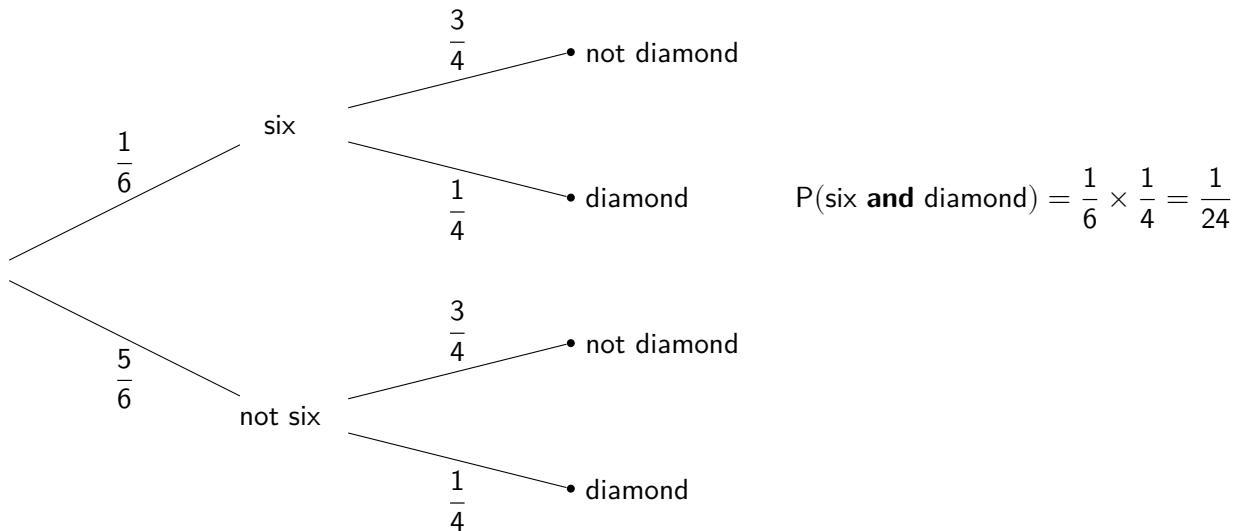
$$P(A \text{ and } B) = P(A) \times P(B)$$

For example, the probability of rolling a die and scoring a five **and** tossing a coin and getting tails is:

$$P(\text{five and tails}) = P(\text{five}) \times P(\text{tails}) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

If knowing that one event has happened *does* change the probability that another event will happen, then those two events are said to be **dependent**. If two events A , B are dependent then the probability of *both* events happening cannot be calculated using the rule above. **Tree diagrams** can be constructed to help calculate probabilities involving two or more events, *whether those events are independent or dependent*.

For example, consider a game involving rolling a six-sided die and drawing a playing card at random from a standard deck of 52 cards, with an interest in the chance of the die landing on a *six* and the card drawn being a *diamond*:



Tree diagrams should be complete with each *branch* labelled with a *probability*, and each *node* labelled with an *event*, as above. Since each split into *sibling branches* should show all the possible things that can happen next, the probabilities on sibling branches should add to 1.

$$\text{i.e. } \frac{1}{6} + \frac{5}{6} = 1 \quad \text{and} \quad \frac{3}{4} + \frac{1}{4} = 1$$

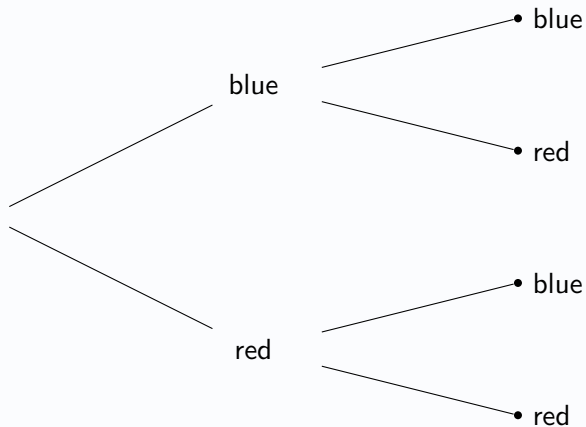
Multiplying the probabilities along a path of branches gives the probability of all those events occurring together.

Note that in the tree diagram on the previous page the probability of the card being a diamond does not change depending on whether the roll of the die gives a six or not; this is because the roll of the die and the drawing of the card are **independent**.

In the following examples the probabilities on the second stage of the tree diagram do change depending on the result of the first stage, since the first and second draws from the bag are **not independent**.

Example 1

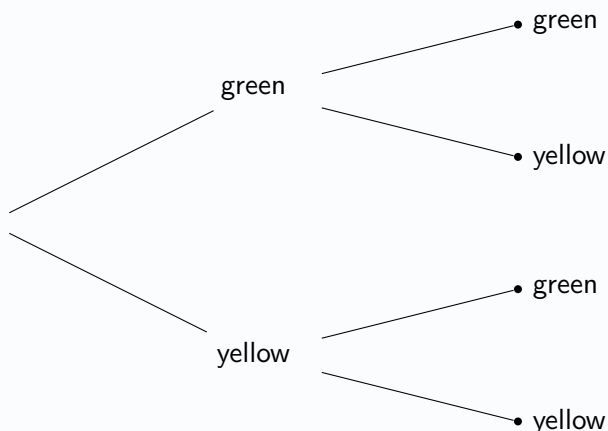
From a bag containing four blue and six red cubes one is drawn at random, followed by another without replacing the first. Find the probability that two red cubes are drawn.



To find all outcomes which satisfy some condition, it may be required to add together several probabilities at the end of the chains.

Example 2

A game involves a player randomly selecting one card then a second card from a set of six green cards and four yellow cards, without replacement. Find the probability of a player selecting at least one yellow card.



Chapter 2 Review

Use this checklist when revising for any assessment/exam. Please ask for help with anything you don't understand.

	Skills & Knowledge	Practice Questions			
2.1 Contingency Tables	I can calculate a basic probability. I can calculate a basic probability from a contingency table. I can calculate and/or probabilities from a contingency table. I can calculate an expected value using a probability.				
2.2 Venn Diagrams	I can complete a two-circle Venn diagram. I can find a probability using a two-circle Venn diagram. I can complete a three-circle Venn diagram. I can find a probability using a three-circle Venn diagram.				
2.3 Tree Diagrams	I can explain what independent and dependent events are. I can calculate the probability of independent events occurring. I can construct a tree diagram. I can calculate probabilities using a tree diagram.				

Notes